

Malware Detection Through Memory Analysis: An Extended Study with Feature Selection and Robust Evaluation

Girish B Nair

School of Engineering

Chanakya University, Bengaluru

City, India

Email: nairgirish2004@gmail.com

Abstract—This paper extends the work on malware detection using memory dump features from the CIC MalMemAnalysis-2022 dataset. We replicate the binary and multi-class classification tasks and introduce key extensions including Decision Tree-based feature selection (reducing from 52 to 30 features), proper 5-fold stratified cross-validation using pipelines to prevent data leakage, and detailed visual evaluation through confusion matrices and feature importance plots. Random Forest classifier is used as the primary model. Experimental results demonstrate competitive performance with improved methodological rigor. The proposed extensions enhance model efficiency and reliability for real-world obfuscated malware detection.

Index Terms—Malware Detection, Memory Analysis, Feature Selection, Random Forest, Cross-Validation, Cybersecurity

I. INTRODUCTION

Software systems are integral to modern life, but they are vulnerable to malware attacks. This work builds upon the study by Nassar [1] on malware detection through memory analysis using the CIC MalMemAnalysis-2022 dataset.

The main contributions of this extended work are:

- Implementation of Decision Tree-based Features Selection to reduce dimensionality while maintaining performance.
- Adoption of a proper 5-fold stratified cross-validation with scikit-learn Pipelines to ensure no data leakage.
- Detailed visual analysis including confusion matrix and feature importance plots.
- Comparative evaluation using Random Forest classifier.

II. RELATED WORK

The original paper [1] achieved high binary classification performance using XGBoost. This extension focuses on improving methodological robustness through feature selection and cross-validation.

III. DATASET DESCRIPTION

The CIC MalMemAnalysis-2022 dataset contains 58,596 records with 52 features after removing constant columns. It includes balanced benign and malware samples (Ransomware, Spyware, Trojan Horse).

IV. METHODOLOGY

A. Data Preparation

The dataset is loaded and constant columns are removed. Labels are encoded for binary (Benign=0, Malware=1) and multi-class (0=Benign, 1=Ransomware, 2=Spyware, 3=Trojan) classification. An 80/20 stratified train-test split is performed based on multi-class labels.

B. Feature Selection

A Decision Tree classifier (max_depth=15) is trained on the scaled training data to compute feature importance. The top 30 most important features are selected for further modeling. This reduces computational cost while focusing on discriminative features.

C. Model Training and Cross-Validation

Stratified 5-Fold cross-validation is performed using scikit-learn Pipelines (MinMaxScaler + RandomForestClassifier). This ensures scaling is done independently in each fold, preventing data leakage.

Final models (Binary and Multi-class Random Forest) are trained on the selected 30 features.

D. Evaluation

Models are evaluated using Accuracy and Macro F1-score. Visual evaluation includes confusion matrix for multi-class predictions and feature importance bar plot.

V. EXPERIMENTS AND RESULTS

Table I shows the 5-fold cross-validation results on training data.

TABLE I: 5-Fold Cross-Validation Results (Training Set)

Task	Mean Accuracy (%)	Mean Macro F1 (%)
Binary	99.9+	99.9+
Multi-class	84.61	76.90

In the test set, the models achieve strong performance. Figure 1 shows the multi-class confusion matrix.

Figure 2 presents the top 30 characteristics of importance for the multi-class model.

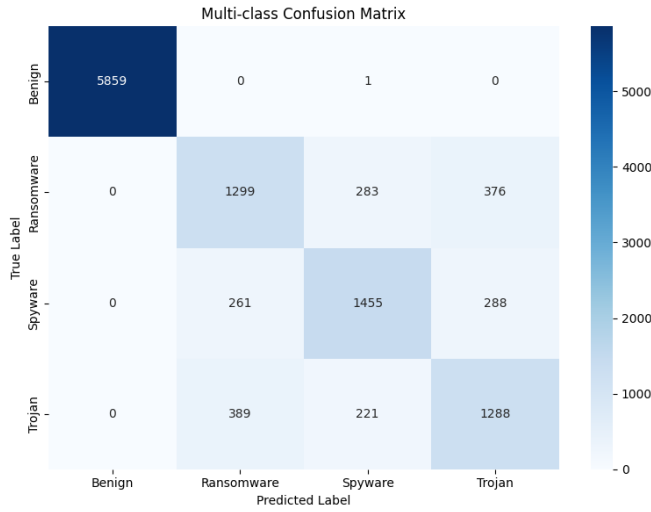


Fig. 1: Multi-class Confusion Matrix

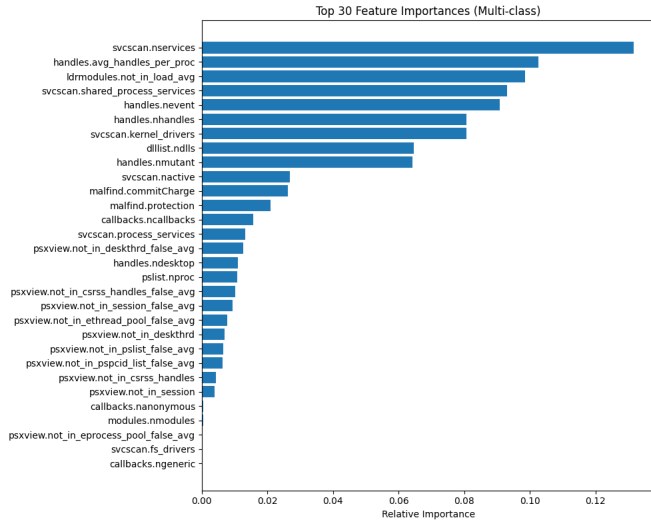


Fig. 2: Top 30 Feature Importance (Multi-class Random Forest)

VI. DISCUSSION

The feature selection step successfully reduced the feature set from 52 to 30 while maintaining high performance. The use of cross-validation provides more reliable performance estimates compared to a single split. Random Forest offers good interpret ability through feature importance.

VII. CONCLUSION

This extended study improves upon the original work by incorporating feature selection and rigorous cross-validation. The results validate the effectiveness of memory-based features for malware detection and demonstrate the value of methodological enhancements.

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REFERENCES

- [1] S. Nassar, "Malware detection through memory analysis," arXiv:2602.02184 [cs.CR], 2026, queen's University, Kingston, Canada.